Does the Brand Survive? Measuring and Enforcing DESIGN.md Adherence in Flowstep's AI-Generated UI

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Abstract

Flowstep lets a team attach a brand's DESIGN.md to a file, and its site says nothing drifts out of style. I measured how much of what Flowstep actually paints lands on the brand's tokens, and built a deterministic fixer for the rest.

Method. Flowstep screens reference theme variables whose values its API does not return. I therefore **recovered each file's theme from Flowstep's own PNGs** with sentinel rendering. With the recovered theme, my renderer matches Flowstep's images to a median 2.2% of pixels. Every painted colour, font size, family, spacing and radius was scored against the tokens: 6,218 values in total. The study was **pre-registered** and sized to the 15-message free trial. It used four brands, 8 briefs with guidelines (G), 4 matched briefs without (N), and 2 drift edits.

Findings.

- **Guidelines work.** Median on-token colour rises from 18% (N) to 51% (G), and the other categories from 74% to 97%. Both tests give $p < 0.005$.
- **Flowstep re-tints the brand.** Its painted primary is darker than the DESIGN.md primary in **8 of 8 files**, and beyond one just-noticeable difference in 6. The same DESIGN.md produced different primaries in different files.
- **Every one of the 24 guided screens** has at least one off-token value.
- **Edits do not drift.** Colour adherence was 54%, 54% and 53% across two edits.
- **One brand font never renders.** Manrope is unavailable, and one file fell back to a Times-like serif.

Repair. `ontoken` snaps values to tokens by editing class strings only. It brought **33 of 36 screens to 100% on-token**, moving a median 6% of pixels, and introduced no contrast failure. Written back into Flowstep in place, 23 of 24 fixed screens were at least as on-brand in Flowstep's own renderer (median 0.993 to 1.000). Two pre-registered checks failed, and both are reported.

AI design tools · DESIGN.md · design tokens · evaluation · Tailwind · MCP · pre-registration

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1. The question

A team that adopts an AI design tool hands over its brand: colours, type, spacing and corner shapes, written as tokens. Flowstep accepts these as a DESIGN.md (Google's open format) attached to a file. It injects them into every generation and promises that "nothing drifts out of style".

That promise is testable, and testing it is the kind of **evaluation-driven development** Flowstep's AI Engineer role asks for. I asked three things:

1. How much of the painted UI is exactly on the tokens, with and without guidelines?
2. Does it stay there through chat edits?
3. Can a cheap, deterministic post-pass close the gap without changing the design?

2. Method

Generation. Everything ran through Flowstep's public MCP server (`api.flowstep.ai/mcp`, OAuth 2.1 PKCE) from a TypeScript client. Every call is logged.

Brands. I wrote four DESIGN.md brands, each passing `@google/design.md lint` with 0 errors:

- **Fjord**, finance: teal, Inter;
- **Ember**, food: burnt orange, DM Sans, off-Tailwind sizes 13/15/17/22/28/34;
- **Moss**, health: green, Manrope, an 8-point scale with no 12 px step;
- **Ink**, editorial: red, Source Serif 4 headlines, sharp corners.

Briefs. Each brief asks for three screens and never mentions colour, type, spacing or shape. G and N receive identical briefs.

Theme recovery. Flowstep's JSX says `bg-primary` and `text-muted-foreground`, not hex values, and the API returns no theme. To find the colours:

1. Render each screen with every theme variable set to a unique sentinel colour.
2. Read what Flowstep painted at each sentinel's pixels; the mode wins.
3. Fit fonts and radius steps by pixel loss.

The calibrated render differs from Flowstep's PNG on a median 2.2% of pixels, against up to 15.7% with a default theme. The worst file is 20%, because I replace photos with grey.

Measurement. Every visible element yields its painted values:

- text, icon, background, border and gradient colours, composited over what is behind them;
- font size and family;
- padding, margin and gap above 2 px;
- radii where they show.

A colour is on-token if it is within $\Delta E_{OK} \leq 0.02$ of a token, which is about one just-noticeable difference. A dimension is on-token within 0.5 px. Text also gets a WCAG AA contrast check.

Repair. Each off-token value snaps to the nearest token:

- colours use chroma-weighted OKLab, so a grey stays a grey, and are the nearest token that keeps AA;
- borders prefer `border`-role tokens;
- inherited colours are fixed where they are declared;
- the pass re-renders and repeats until stable, at most 4 passes.

Only `className` strings change, merged with tailwind-merge, and every other byte of the JSX is preserved.

Pre-registration. The protocol, metrics, repair rules and hypotheses were frozen in commit `af1b7d0`, with SHA-256 pins, before any confirmatory generation. The free trial allowed 15 messages, and the design used them as follows:

Part	Files	Messages
Development	D1 (G)	1
Paired	F1, E1, M1, I1 × {G, N}	8
G only	F2, E2, M2, I2	4
Drift	F1-G, two content-only edits	2

3. Results

Table 1 gives every pre-registered verdict, frozen and amended (§3.6).

	Hypothesis	Frozen	Amended
H1a	colour G > N	✓ p=.002	✓ p=.005
H1b	non-colour G > N	✓ p=.0002	✓ p=.0005
H2	primary shifted (≥75% of files)	✓ 6/8	✓ 6/8
H3	off-token values remain (≥75% of G screens)	✓ 24/24	✓ 24/24
H4a	≥90% of screens fully repaired	✓ 33/36	✓ 33/36
H4b	AA failures down ≥80%	✓ 205→2	**✗ 8→2**
H4c	no new AA failure	✓ 0	✓ 0
H4d	median pixels moved ≤25%	✓ 7.8%	✓ 6.0%
H5	round trip via `add-screen-jsx`	**✗ 18/24**	**✗ 18/24**
H6	renderer median diff ≤3%	✓ 2.2%	✓ 2.2%

Table 1. Pre-registered verdicts. Wilcoxon signed-rank tests are exact and one-sided, on 12 screen pairs.

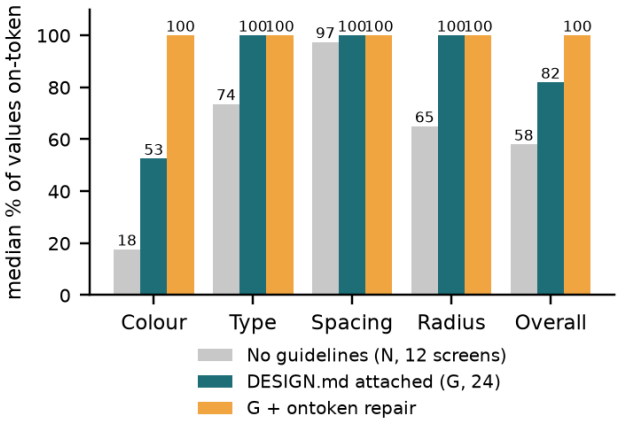


Figure 1. Median share of painted values on the brand's tokens, by category, over all 12 N and 24 G screens (amended analysis). H1 uses the 12 matched pairs, hence 51% there against 53% here.

3.1 Guidelines help, a lot

Attaching the DESIGN.md roughly triples exact-colour use, from a median of 18% to 51%. Type, spacing and radius together rise from 74% to 97%.

Spacing is near-perfect even without guidelines (97%), because Tailwind's 4 px grid already matches most brand scales. Type is where guidelines matter most outside colour. Ember's off-Tailwind sizes were still missed in one file of two (E1-G: 42% of sizes on-token; E2-G: 100%).

3.2 Flowstep re-tints the brand

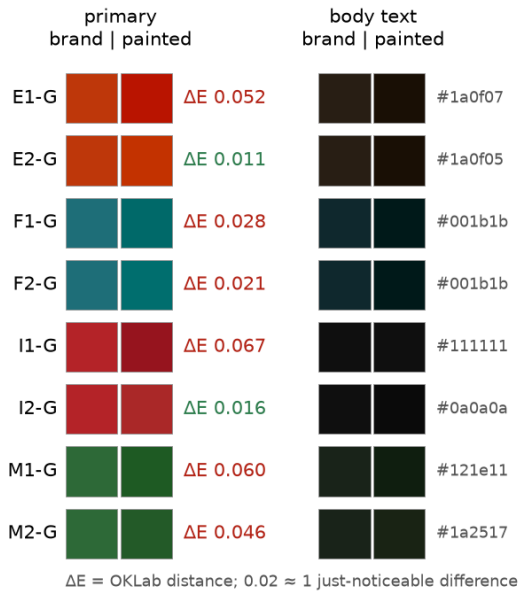


Figure 2. The brand's primary and body-text colours against what Flowstep painted, for every guided file.

In every one of the 8 guided files, the painted primary is **darker** than the token: ΔL_{OK} from −0.005 to −0.064. In 6 files the shift exceeds one just-noticeable difference, with ΔE up to 0.067 (Ink: #B4232A painted as #97181F).

The same DESIGN.md yields **different** primaries in different files: Ember's #C23B0B came out as #B91700 in

one file and #C43300 in another. The theme is evidently re-derived per generation rather than copied from the tokens.

Body text shows the same darkening: Fjord's `ink` #0F2A2E is painted #001B1B. Radius steps are re-scaled too. On the development file, `rounded-xl` painted 6 px where Flowstep's own internal prompt asked for 12 px.

3.3 Nothing drifts across edits

Two content-only edits on F1-G ("add a promotional banner", "add a secondary action") left the recovered theme identical, and left median colour adherence at 54%, 54% and 53%. On this sample, Flowstep's no-drift claim holds. With n = 1 file this is descriptive only.

3.4 One brand font never renders

Flowstep's PNG renderer painted Inter, DM Sans and Source Serif 4 correctly. **Manrope never rendered.**

M1-G writes `font-[Manrope]` with no fallback stack, so Flowstep paints a Times-like default serif. M2-G silently used Inter instead. A brand whose font is not in the renderer's set gets no warning. This finding is exploratory.

3.5 The repair

On the calibrated renderer, 33 of 36 screens reach 100% on-token with zero unrepairable values. The median screen moves 6% of its pixels, and almost none of them visibly: the median pixelmatch difference is 0.07%. No screen gains a contrast failure.

The 3 incomplete screens are all unguided and show two gaps in the repair:

- text faded with `opacity-70` stays faint after its colour is fixed;
- progress rings drawn as arbitrary `conic-gradient(…)` values are not rewritten.

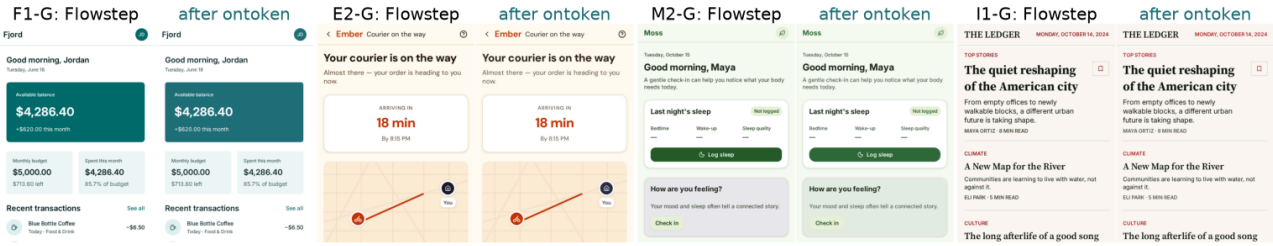


Figure 3. Flowstep's own render of four guided screens, original and after ontoken's repair is written back in place (amended repair). Layout, copy and fonts are untouched; colours and radii move onto the tokens.

Round trip. H5 failed: only 18 of 24 screens added back with `add-screen-jsx` stayed at least as on-brand. The cause is structural. Flowstep stores the theme on each AI-generated screen, and a code-added screen starts from the default theme. Every on-token colour the repair left as a theme variable then turned white or black (Ember lost its cream and orange).

Writing the same JSX over the original screen with `replace-screen-jsx` keeps the theme. Then **23 of 24 screens** are at least as on-brand in Flowstep's own renderer, with the median flat-pixel brand adherence going from 0.993 to 1.000. The single "worse" screen moved by 0.000001, from 0.999769 to 0.999768.

3.6 What went wrong in my pipeline

Two bugs surfaced after the test run. Both are fixed in separate files, with the frozen code still pinned.

Text colours. Theme variables used only for text were recovered from thin glyph pixels, where the background often won. For example, Ink's `--foreground` came out as the paper colour. That produced 197 false contrast failures. The fix re-votes text variables with background pixels excluded, and now recovers Ink's text as #111111, the exact token. The fix turns H4b from pass to fail.

Fonts. The frozen repair wrote Tailwind v4's `font-[family-name:X]`, which Flowstep ignores, and inserted Manrope, which Flowstep cannot paint. In Flowstep, this turned Ink's serif headlines sans and Moss's text into Times. The amended repair writes `font-['X']` and never inserts a font Flowstep lacks.

4. What this suggests for Flowstep

1. **Copy tokens, don't re-derive them.** Map DESIGN.md colours to theme variables as given, and check the mapping with an automatic ΔE test. It is cheap and would have caught 6 of 8 files here.
2. **Make adherence an eval in CI.** `ontoken check` scores a screen in under 4 s from a cold start, and about 0.3 s per render in batch. Run it on a fixed prompt set per model or prompt change and track per-category rates.
3. **Load or reject brand fonts.** Use fallback stacks in emitted classes, and warn when a DESIGN.md font is unavailable.
4. **Let code-first screens inherit a theme.** `add-screen-jsx` could accept a theme or take the file's last one.
5. **Ship the snap pass as a guard.** It is deterministic, preserves layout and copy, and costs no model call.

5. Limitations

- **Sample size.** It is small: 36 screens and 6,218 values from 15 messages on one day and one model version.
- **Brands.** The four brands are my own. Real brands may define larger palettes and allow tints.

- **Renderer.** Mine is calibrated, not Flowstep's. Photos are replaced with grey, and recovered colours are modes over pixels.
- **Fonts.** The font findings rest on visual and pixel evidence from Flowstep's PNG export. The live editor may load fonts differently.
- **Amendments.** Two were made after the test run (§3.6). Both verdict sets are reported, and the frozen code is pinned.

6. Reproduce

```
npm ci && npx playwright install chromium
npx vitest run                     # 30 tests
npx tsx study/measure.ts --v2      # offline
npx tsx study/analyze.ts --v2
npx tsx src/cli.ts check screen.jsx --design brand.md
```

Fresh generations need a Flowstep account: `npx tsx study/run.ts --phase test`.